Fine-Structure Constant: Unit Conventions Why $\alpha = 1$ can be set

Supplement to Document 011

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Abstract

This document addresses aspects of the fine-structure constant not discussed in detail in Document 011. The focus is on providing a comprehensive justification for why and how $\alpha = 1$ can be set (Heaviside-Lorentz convention), the physical consequences of different unit systems, and the historical and practical implications of redefining electromagnetic units.

For T0-specific derivations (characteristic energy E_0, geometric parameter ξ, T0 formula $\alpha = \xi(E_0/1\text{MeV})^2$) see Document 011.

.1 Introduction and Reference to Document 011

Delimitation from Document 011

Document 011 covers in detail:

- T0 derivation: $\alpha = \xi(E_0/1\text{MeV})^2$
- Characteristic energy: $E_0 = \sqrt{m_e \cdot m_\mu} = 7.348$ MeV (bare), corrected $E_0 = 7.348/\sqrt{K_{\text{frak}}} = 7.398$ MeV
- Geometric parameter: $\xi = \frac{4}{3} \times 10^{-4}$

- Alternative formulations: with μ_0, with r_e/λ_C, etc.
- Historical context (Sommerfeld)
- Natural units and energy as fundamental field
- Detailed dimensional analysis of all formulations

This document (044) focuses on:

- **Why** $\alpha = 1$ can be set (detailed justification)
- **How** different unit conventions work
- Consequences of redefining the Coulomb
- Heaviside-Lorentz vs. Gauss vs. SI units
- Practical aspects and historical development
- Fine's inequality vs. fine-structure constant (name confusion)

Why Two Documents?

Document 011: T0 theory and physical derivations
Document 044: Unit systems and conventions
Both complement each other, with minimal overlap.

.2 Different Unit Conventions for α

Overview of Systems

The fine-structure constant can be expressed in different unit systems:

System	Formula	Value
SI Standard	$\alpha = \frac{e^2}{4\pi\varepsilon_0\hbar c}$	$\approx \frac{1}{137}$
Heaviside-Lorentz	$\alpha = \frac{e^2}{4\pi}$ (with $\hbar = c = \varepsilon_0 = 1$)	1 or $\frac{1}{137}$
Gauss (cgs)	$\alpha = \frac{e^2}{\hbar c}$	$\approx \frac{1}{137}$

Table 1: Unit systems for α

Important: The numerical value depends on the convention, the *physical* predictions do not.

.3 Heaviside-Lorentz Units in Detail

What are Heaviside-Lorentz Units?

The Heaviside-Lorentz system is a variant of natural units, specifically for electrodynamics:

$$\boxed{\hbar = c = \varepsilon_0 = 1} \tag{1}$$

Consequences:

- Electromagnetic equations become more symmetric
- The factor 4π disappears from many formulas
- Elementary charge is redefined

Why $\varepsilon_0 = 1$?

In SI units, 4π appears in many electromagnetic formulas:

$$\vec{E} = \frac{1}{4\pi\varepsilon_0}\frac{q}{r^2}\hat{r} \quad \text{(Coulomb's law)} \tag{2}$$

$$\nabla \cdot \vec{E} = \frac{\rho}{\varepsilon_0} \quad \text{(Maxwell's equation)} \tag{3}$$

With $\varepsilon_0 = 1$, these become:

$$\vec{E} = \frac{q}{4\pi r^2}\hat{r} \tag{4}$$

$$\nabla \cdot \vec{E} = \rho \tag{5}$$

The factor 4π remains only in Coulomb's law (sphere surface $4\pi r^2$); Maxwell's equations become free of 4π.

Fine-Structure Constant in Heaviside-Lorentz

Starting point (SI):

$$\alpha = \frac{e^2}{4\pi\varepsilon_0\hbar c} \tag{6}$$

With $\hbar = c = \varepsilon_0 = 1$:

$$\alpha = \frac{e^2}{4\pi \cdot 1 \cdot 1 \cdot 1} = \frac{e^2}{4\pi} \tag{7}$$

Now the crucial question: What value does e have in this system?

.4 Two Variants of Heaviside-Lorentz

Variant A: Normalize e so that $\alpha = 1$

Approach: We define the unit of charge such that $\alpha = 1$.
Since $\alpha = e^2/(4\pi)$ in HL units:

$$e^2 = 4\pi \quad \Rightarrow \quad e = \sqrt{4\pi} \approx 3.545 \tag{8}$$

Physical meaning:

- Elementary charge becomes a *dimensionless unit*
- Electromagnetic coupling is "normalized"
- Charge is measured in units of $\sqrt{\hbar c}$

What changes?
The elementary charge gets a new numerical value:

$$e_{\text{HL}} = \sqrt{4\pi\varepsilon_0\hbar c} \quad \text{(expressed in SI units)} \tag{9}$$

Numerically:

$$e_{\text{HL}} = \sqrt{4\pi \times 8.854 \times 10^{-12} \times 1.055 \times 10^{-34} \times 3 \times 10^{8}} \tag{10}$$

$$\approx 1.877 \times 10^{-18}\ \text{C} \tag{11}$$

This is $\sqrt{137.036} \times e_{\text{SI}}$ – the charge for which $\alpha = 1$. In the HL charge unit $\sqrt{\varepsilon_0\hbar c} \approx 5.29 \times 10^{-19}$ C it has the numerical value $\sqrt{4\pi}$.

Variant B: Keep e, $\alpha \approx 1/137$

Approach: The elementary charge retains its "natural" value.
In this case:

$$\alpha = \frac{e^2}{4\pi} \approx \frac{1}{137} \tag{12}$$

because e in these units has the value $\sqrt{4\pi/137} \approx 0.303$.
Physical meaning:

- Charge retains physical meaning
- α remains $\approx 1/137$
- Only the mathematical form simplifies

Which variant is used?

In practice:

- **T0 theory:** Sets all constants = 1 ($c = \hbar = \alpha = G = 1$)
- **Theoretical high-energy physics:** Often $\hbar = c = 1$, sometimes also $\alpha = 1$
- **Numerical calculations:** Often $\hbar = c = 1$, but $\alpha \approx 1/137$
- **Experimental physics:** Almost always SI units (all constants have numerical values)
T0 convention:
- In T0 calculations: $c = \hbar = \alpha = G = 1$ (maximum simplification)
- Only parameter: $\xi = \frac{4}{3} \times 10^{-4}$ (integer structural choices are motivated settings; SI anchors such as m_e serve for conversion)
- When comparing with experiments: SI values ($c = 3 \times 10^8$ m/s, $\alpha \approx 1/137$, etc.)
- Both describe the same physics.

.5 Reconstruction of SI Values from T0

The Central Principle

Basic idea: Although T0 sets all constants to 1, the SI values can be reconstructed.

T0 Reconstruction

In T0 calculations:

- All constants = 1: $c = \hbar = \alpha = G = 1$
- Only parameter: $\xi = \frac{4}{3} \times 10^{-4}$ (plus structural choices as settings)
- Formulas maximally simplified

Reconstruction of SI values:

- Fine-structure constant: $\alpha_{\text{SI}} = \xi(E_0/1\text{MeV})^2 \approx 1/137$
- Gravitational constant: $G_{\text{SI}} = \frac{\xi^2}{4m_e} \times$ factors
- Further constants: reconstructable from ξ and energy scales (see Doc. 011 and 012)

Example: Fine-Structure Constant

In T0 units:

$$\alpha = 1 \tag{13}$$

Reconstruction of SI value:

$$\alpha_{\text{SI}} = \xi \cdot \left(\frac{E_0}{1\,\text{MeV}} \right)^2 \tag{14}$$

With $\xi = \frac{4}{3} \times 10^{-4}$ and $E_0 = 7.398$ MeV:

$$\alpha_{\text{SI}} = 1.3333 \times 10^{-4} \times (7.398)^2 \tag{15}$$

$$= 1.3333 \times 10^{-4} \times 54.73 \tag{16}$$

$$= 7.297 \times 10^{-3} \tag{17}$$

$$= \frac{1}{137.04} \tag{18}$$

Experimental: $\alpha_{\text{exp}} = \frac{1}{137.036}$

Deviation from the experimental value: 0.0005 % (unrounded $1/(1.3333 \times 10^{-4} \times 54.730) = 1/137.035$); even the rounded value 137.04 deviates by only $0.003\,\%$.

Example: Gravitational Constant

In T0 units:

$$G = 1 \tag{19}$$

Reconstruction of SI value:

$$G_{\text{SI}} = \frac{\xi^2}{4m_e} \times C_{\text{conv}} \times K_{\text{frak}} \tag{20}$$

where:

- C_{conv} = Conversion factors (eV → J, etc.)
- K_{frak} = fractal correction

A separate dimension conversion $C_{\text{dim}} = 1/E_{\text{char}} = 3.472 \times 10^{-2}$ does not appear: it is cancelled again by the factor $E_{\text{char}} = 28.8$ in the step chain of Doc. 012; inserted into the formula on its own it would give 2.35×10^{-12} instead of 6.674×10^{-11}.

Detailed derivation see Document 012 (Gravitation).

Why does this work?

Key: ξ is dimensionless and is taken as universal in the T0 framework.

1. In T0: ξ determines all coupling strengths
2. In SI: ξ together with characteristic energies (E_0, masses) reconstructs all constants
3. Physical predictions: identical in both systems
4. Only the mathematical representation differs

Constant	T0	SI	Reconstruction
c	1	3×10^8 m/s	Convention
$\hbar$	1	1.055×10^{-34} J·s	Convention
α	1	$\approx 1/137$	$\xi(E_0/1\text{MeV})^2$
G	1	6.67×10^{-11} m³/(kg·s²)	$\xi^2/(4m_e)\times$ factors
ξ	$\frac{4}{3} \times 10^{-4}$	$\frac{4}{3} \times 10^{-4}$	same

Table 2: T0 vs. SI - Reconstruction of constants

Comparison Table

Important Conclusion

- **T0 is not new physics**, but a reparametrization
- Instead of many constants (c, $\hbar$, α, G, ...) only **one parameter** ξ
- All SI values reconstructable from ξ and energy scales
- Advantage: Formulas simpler, physical relationships clearer
- Disadvantage: Conversion to SI needed for experiments

.6 Why can $\alpha = 1$ be set?

Fundamental Insight

Core Statement

The fine-structure constant α is a **dimensionless number**. It contains two levels that must be kept apart. The number "137" as the name of the coupling depends on the **normalisation** (charge unit, 4π, $\hbar$) and is a choice. The **scale ratio** it expresses, for example r_e/λ_C or binding energy relative to rest energy in hydrogen, is physical and the same in every system of units (Doc. 386).
One can set $\alpha = 1$ by redefining the **unit of charge** accordingly.

Step-by-Step Justification

Step 1: What is a convention?
SI units are historically evolved definitions:

- 1 meter = originally 1/10,000,000 of the Earth's meridian
- 1 second = originally 1/86,400 of a solar day
- 1 Coulomb = defined via Ampere and force between currents

None of these is "fundamental".
Step 2: α in SI
In SI units:

$$\alpha = \frac{e^2}{4\pi\varepsilon_0\hbar c} \approx \frac{1}{137} \tag{21}$$

Because α is dimensionless, the value $1/137$ does not depend on how the Coulomb or ε_0 was defined; it comes out the same in every system of units, as long as the coupling is normalised with 4π and $\hbar$. With h instead of $\hbar$ the same coupling would read $2\alpha \approx 1/68.5$. What is fixed by choice is the normalisation, not the ratio.

Step 3: Redefinition

We can say: "From now on, the elementary charge is no longer 1.602×10^{-19} C, but $e = \sqrt{4\pi\varepsilon_0\hbar c}$."

Then automatically:

$$\alpha = \frac{(\sqrt{4\pi\varepsilon_0\hbar c})^2}{4\pi\varepsilon_0\hbar c} = 1 \tag{22}$$

Step 4: Physical Consequences

- **No physical prediction changes.**
- Only the *numbers* in formulas change
- All ratios remain the same
- All experiments yield the same results

Analogy: Temperature Scales

Celsius: Water freezes at 0°C

Fahrenheit: Water freezes at 32°F

Kelvin: Water freezes at 273.15 K

None of these scales is more "correct" than the others; they are conventions.

Similarly, the choice of charge unit that makes $\alpha = 1$ is a choice of convention. The scale ratio does not disappear; it then sits in the conversion between the historical and the new charge unit (Doc. 386).

.7 Consequences of Redefining the Coulomb

What does it mean to redefine elementary charge?

If e is redefined such that $\alpha = 1$:

Old definition (SI):

$$e = 1.602 \times 10^{-19} \text{ C} \tag{23}$$

New definition (HL with $\alpha = 1$):

$$e = \sqrt{4\pi} \approx 3.545 \quad \text{(dimensionless in natural units)} \tag{24}$$

or expressed in SI units:

$$e_{\text{new}} = \sqrt{4\pi\varepsilon_0\hbar c} \approx 1.877 \times 10^{-18} \text{ C} = \sqrt{137.036}\, e \tag{25}$$

The charge unit is rescaled by the same factor: $1\,\text{C}_{\text{new}} = (e_{\text{new}}/e) \cdot 1\,\text{C} = \sqrt{137} \times 1\,\text{C}_{\text{old}}$. Numerical values transform inversely to the units: a unit larger by a factor f gives a numerical value smaller by f. (Here f is the scaling factor between the old and the new unit, not a frequency and not the base winding number $f = 1/\xi$.)

Effects on Electromagnetic Quantities

Electric Current (Ampere)

Since $1\ \text{A} = 1\ \text{C/s}$:

$$1\ \text{A}_{\text{new}} = \frac{1\ \text{C}_{\text{new}}}{1\ \text{s}} = \sqrt{137} \times 1\ \text{A}_{\text{old}} \tag{26}$$

Electric Voltage (Volt)

$1\ \mathrm{V} = 1\ \mathrm{J/C}$:

$$1\ \mathrm{V}_{\mathrm{new}} = \frac{1\ \mathrm{J}}{1\ \mathrm{C}_{\mathrm{new}}} = \frac{1}{\sqrt{137}} \times 1\ \mathrm{V}_{\mathrm{old}} \tag{27}$$

Capacitance (Farad)

$$1\ \mathrm{F}_{\mathrm{new}} = \frac{1\ \mathrm{C}_{\mathrm{new}}}{1\ \mathrm{V}_{\mathrm{new}}} = 137 \times 1\ \mathrm{F}_{\mathrm{old}} \tag{28}$$

Are these changes "real"?

No. They are conversion factors, like Celsius → Fahrenheit.

All physical ratios remain identical:

- Capacitance of a capacitor / distance: same
- Force between charges / distance²: same
- All experiments: same results

Only the *numerical values* we calculate with change.

.8 Practical Impacts on Everyday Calculations

Motivation

Question: If we set $\alpha = 1$, what does that mean for ordinary electrical calculations with volts, amperes, resistance, capacitance?

Answer: All formulas change, but the *physical results* remain identical.

Example 1: Ohm's Law

In SI Units (Standard)

$$U = R \cdot I \tag{29}$$

Numerical example:

- Resistance: $R = 100\ \Omega$
- Current: $I = 2$ A
- Voltage: $U = 100 \times 2 = 200$ V

In Heaviside-Lorentz with $\alpha = 1$

The formula remains $U = R \cdot I$, but the *numerical values* change.

Unit conversion:

$$1\ \mathrm{A}_{\mathrm{new}} = \sqrt{137} \times 1\ \mathrm{A}_{\mathrm{old}} \approx 11.7\ \mathrm{A}_{\mathrm{old}} \tag{30}$$

$$1\ \mathrm{V}_{\mathrm{new}} = \frac{1}{\sqrt{137}} \times 1\ \mathrm{V}_{\mathrm{old}} \approx 0.085\ \mathrm{V}_{\mathrm{old}} \tag{31}$$

$$1\ \Omega_{\mathrm{new}} = \frac{1}{137} \times 1\ \Omega_{\mathrm{old}} \tag{32}$$

Same circuit in new units:

$$R_{\text{new}} = 100 \times 137 = 13\,700\ \Omega_{\text{new}} \tag{33}$$

$$I_{\text{new}} = \frac{2}{\sqrt{137}} \approx 0.171\ \text{A}_{\text{new}} \tag{34}$$

$$U_{\text{new}} = 13\,700 \times 0.171 = 2341\ \text{V}_{\text{new}} \tag{35}$$

Conversion back to SI:

$$U = 2341 \times 0.0854\ \text{V}_{\text{old}} = 200\ \text{V} \quad \checkmark \tag{36}$$

The result is identical.

Example 2: Power of a Light Bulb

In SI Units

$$P = U \cdot I = \frac{U^2}{R} \tag{37}$$

Light bulb: 60 W at 230 V

$$R = \frac{U^2}{P} = \frac{(230)^2}{60} = 882\ \Omega \tag{38}$$

$$I = \frac{P}{U} = \frac{60}{230} = 0.26\ \text{A} \tag{39}$$

In Heaviside-Lorentz with $\alpha = 1$

Power: $1\ \text{W}_{\text{new}} = 1\ \text{W}_{\text{old}}$ (energy/time does not change in HL)

$$U_{\text{new}} = 230 \times \sqrt{137} = 2692\ \text{V}_{\text{new}} \tag{40}$$

$$R_{\text{new}} = 882 \times 137 = 1.21 \times 10^5\ \Omega_{\text{new}} \tag{41}$$

$$I_{\text{new}} = \frac{P}{U_{\text{new}}} = \frac{60}{2692} = 0.0223\ \text{A}_{\text{new}} \tag{42}$$

Verification:

$$P = U_{\text{new}} \cdot I_{\text{new}} = 2692 \times 0.0223 = 60\ \text{W} \quad \checkmark \tag{43}$$

Example 3: Charging a Capacitor

In SI Units

$$Q = C \cdot U \tag{44}$$

Capacitor: $C = 100\ \mu\text{F}$ at $U = 12$ V

$$Q = 100 \times 10^{-6} \times 12 = 1.2 \times 10^{-3}\ \text{C} \tag{45}$$

Stored energy:

$$E = \frac{1}{2}CU^2 = \frac{1}{2} \times 100 \times 10^{-6} \times 144 = 7.2 \times 10^{-3}\ \text{J} \tag{46}$$

In Heaviside-Lorentz with $\alpha = 1$

Conversion:

$$1\ \mathrm{F}_{\text{new}} = 137 \times 1\ \mathrm{F}_{\text{old}} \tag{47}$$

$$1\ \mathrm{C}_{\text{new}} = \sqrt{137} \times 1\ \mathrm{C}_{\text{old}} \tag{48}$$

$$C_{\text{new}} = \frac{100 \times 10^{-6}}{137} = 7.30 \times 10^{-7}\ \mathrm{F}_{\text{new}} \tag{49}$$

$$U_{\text{new}} = 12 \times \sqrt{137} = 140.5\ \mathrm{V}_{\text{new}} \tag{50}$$

$$Q_{\text{new}} = 7.30 \times 10^{-7} \times 140.5 = 1.025 \times 10^{-4}\ \mathrm{C}_{\text{new}} \tag{51}$$

Conversion back:

$$Q = 1.025 \times 10^{-4} \times \sqrt{137} = 1.2 \times 10^{-3}\ \mathrm{C}_{\text{old}} \quad \checkmark \tag{52}$$

Energy:

$$E_{\text{new}} = \frac{1}{2} \times 7.30 \times 10^{-7} \times (140.5)^2 = 7.2 \times 10^{-3}\ \mathrm{J} \quad \checkmark \tag{53}$$

The energy is the same in all systems.

Example 4: RC Time Constant

In SI Units

$$\tau = R \cdot C \tag{54}$$

Circuit: $R = 1\ \mathrm{k}\Omega$, $C = 10\ \mu\mathrm{F}$

$$\tau = 1000 \times 10 \times 10^{-6} = 0.01\ \mathrm{s} = 10\ \mathrm{ms} \tag{55}$$

In Heaviside-Lorentz with $\alpha = 1$

$$R_{\text{new}} = 1000 \times 137 = 1.37 \times 10^{5}\ \Omega_{\text{new}} \tag{56}$$

$$C_{\text{new}} = \frac{10 \times 10^{-6}}{137} = 7.30 \times 10^{-8}\ \mathrm{F}_{\text{new}} \tag{57}$$

$$\tau_{\text{new}} = 1.37 \times 10^{5} \times 7.30 \times 10^{-8} = 0.01\ \mathrm{s} = 10\ \mathrm{ms} \quad \checkmark \tag{58}$$

Time remains the same: physical timescales do not change.

Quantity	SI	HL Factor	HL ($\alpha = 1$)
Charge (Q)	C	$\sqrt{137}$	$\sqrt{137}$ C
Current (I)	A	$\sqrt{137}$	$\sqrt{137}$ A
Voltage (U)	V	$1/\sqrt{137}$	V$/\sqrt{137}$
Resistance (R)	Ω	$1/137$	$\Omega/137$
Capacitance (C)	F	137	137 F
Power (P)	W	1	W (unchanged)
Energy (E)	J	1	J (unchanged)
Time (τ)	s	1	s (unchanged)

Table 3: Conversion factors SI → HL with $\alpha = 1$ (factors for the units; numerical values are divided by the factor)

Summary of Practical Calculations

Important Insights

Core Statement

What changes:
- Numerical values for charge, current, voltage, resistance, capacitance

What does NOT change:
- Energy
- Power
- Time
- All physical ratios
- All experimental results

Conclusion: It is a conversion, comparable to meters ↔ feet.

Why does nobody use $\alpha = 1$ in practice?

Reasons:

1. **Measuring devices:** All voltmeters, ammeters, etc. are calibrated in SI
2. **Standards:** Worldwide accepted SI definitions
3. **Intuition:** Engineers know typical values in SI
   - Household: 230 V, not 2692 V_{new}
   - USB: 5 V, not 58.5 V_{new}
4. **Conversion is laborious:** $\sqrt{137}$ factors everywhere
5. **No advantage for practitioners:** Simplification only visible in theoretical formulas
   But: In theoretical calculations (QED, Feynman diagrams), $\alpha = 1$ can be helpful.

.9 Practical Aspects of Different Systems

Advantages and Disadvantages: SI Units

Advantages:
- Worldwide standardized

- Directly usable for experiments
- All measuring devices calibrated in SI
- Clear separation of length/time/mass/charge

Disadvantages:

- Many constants in formulas ($4\pi\varepsilon_0$, $\hbar$, c)
- Physical relationships obscured
- Dimensions unwieldy

Advantages and Disadvantages: Heaviside-Lorentz with $\alpha = 1$

Advantages:

- Maximally simplified formulas
- Electromagnetic symmetry visible
- Theoretical calculations simpler
- QED Feynman diagrams easier to read

Disadvantages:

- No direct connection to experiments
- Conversion to SI laborious
- Unfamiliar for practitioners
- Physical "size" of e unclear

Advantages and Disadvantages: Natural Units with $\alpha \approx 1/137$

Advantages:

- Simplified formulas ($\hbar = c = 1$)
- α retains physical meaning
- Good compromise theory/practice
- Numerically: $\alpha \ll 1$ → perturbation theory

Disadvantages:

- Still need conversion to SI
- Factor 4π remains in some formulas

  This is the usual convention in present-day particle physics.

.10 Historical Development

Gauss Units (cgs)

19th century: Gauss system (centimeter-gram-second)

$$\alpha = \frac{e^2}{\hbar c} \tag{59}$$

No $4\pi\varepsilon_0$, because $4\pi\varepsilon_0 = 1$ by definition in cgs.

SI Units (MKSA)

20th century: SI system (meter-kilogram-second-ampere)

$$\alpha = \frac{e^2}{4\pi\varepsilon_0 \hbar c} \tag{60}$$

The $4\pi\varepsilon_0$ appears because the SI ampere is defined via force.

Heaviside-Lorentz

Theoretical physics: Heaviside-Lorentz simplifies Maxwell's equations

$$\nabla \times \vec{E} = -\frac{\partial \vec{B}}{\partial t}, \quad \nabla \times \vec{B} = \vec{j} + \frac{\partial \vec{E}}{\partial t} \tag{61}$$

The equations are symmetric (in SI, μ_0 and ε_0 appear asymmetrically)

Natural Units

Modern high-energy physics: $\hbar = c = 1$, but different conventions for α

.11 Fine's Inequality vs. Fine-Structure Constant

Frequent Confusion

Note: *Fine's inequality* and the *fine-structure constant* are different concepts.

Fine's Inequality

What it is:

- A form of Bell's inequality
- Test for local hidden variables
- Quantum entanglement vs. classical correlations

Mathematically:

$$|C(\alpha, \beta) - C(\alpha, \beta')| + |C(\alpha', \beta) + C(\alpha', \beta')| \leq 2 \tag{62}$$

where C are correlation functions.
Physically: Shows non-locality of quantum mechanics

Fine-Structure Constant

What it is:

- Fundamental physical constant
- Strength of the electromagnetic interaction
- Dimensionless, $\alpha \approx 1/137$

Mathematically:

$$\alpha = \frac{e^2}{4\pi\varepsilon_0 \hbar c} \tag{63}$$

Physically: Determines EM coupling strength

No Conceptual Connection

The similarity in name is coincidental; the two concepts are not related.

.12 Summary

Core Statements

1. The number “137” as the name of the coupling depends on the normalisation; the scale ratio that α expresses is physical (Doc. 386)
2. One **can** set $\alpha = 1$ by redefining the unit of charge
3. **T0 theory:** Sets all constants = 1: $c = \hbar = \alpha = G = 1$
4. Only parameter in T0: $\xi = \frac{4}{3} \times 10^{-4}$ (integer structural choices are settings, SI anchors serve for conversion)
5. **No** physical prediction changes
6. Only numerical values in formulas differ
7. When comparing with experiments: SI values ($\alpha \approx 1/137$, $c = 3 \times 10^8$ m/s, etc.)

For Further Details See Document 011

- T0 derivation of α
- Characteristic energy E_0
- Geometric parameter ξ
- Experimental verification
- Detailed dimensional analysis
- Historical context (Sommerfeld)

0.1 Conversion Table: SI ↔ Heaviside-Lorentz

Quantity	SI	HL ($\alpha = 1$)
Elementary charge	$e = 1.602 \times 10^{-19}$ C	$e = \sqrt{4\pi} \approx 3.545$
Fine-structure constant	$\alpha \approx 1/137$	$\alpha = 1$
ε_0	8.854×10^{-12} F/m	1
$\hbar$	1.055×10^{-34} J·s	1
c	3×10^8 m/s	1

Table 4: Conversion table SI to HL

0.2 Sample Calculation: Coulomb’s Law

In SI Units

$$F = \frac{1}{4\pi\varepsilon_0} \frac{e^2}{r^2} \tag{64}$$

Numerically for $r = 1$ Å = 10^{-10} m:

$$F = \frac{1}{4\pi \times 8.854 \times 10^{-12}} \frac{(1.602 \times 10^{-19})^2}{(10^{-10})^2} \tag{65}$$

$$\approx 2.3 \times 10^{-8} \text{ N} \tag{66}$$

In HL Units ($\hbar = c = \varepsilon_0 = 1$)

$$F = \frac{e^2}{4\pi r^2} = \frac{\alpha}{r^2} \tag{67}$$

With r in natural units: $r = 1$ Å $= 5.07 \times 10^{-4}$ eV^{-1} (with $\hbar c = 1.973 \times 10^{-7}$ eV m)

$$F = \frac{\alpha}{(5.07 \times 10^{-4})^2} = \frac{3.89 \times 10^6}{137.036} \approx 2.84 \times 10^4 \text{ eV}^2 \tag{68}$$

Conversion to SI: 1 eV$^2 = (1\,\text{eV})^2/(\hbar c) \approx 8.12 \times 10^{-13}$ N

$$F \approx 2.3 \times 10^{-8} \text{ N} \tag{69}$$

The result is identical; only the intermediate steps look different.

The SI result 2.3×10^{-8} N follows only with the numerical value $\alpha = 1/137.036$ in the numerator; with $\alpha = 1$ one would get 3.2×10^{-6} N. The force depends on the physical value of α, not on the choice of charge unit.